Project Proposal: Library Management System

By

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1. Introduction

Libraries remain a central resource for research, learning, and knowledge access within academic institutions. However, the school library still relies on manual, paper-based processes to manage their collections, member records, and lending activities. This project proposes the design and implementation of a **Library Management System (LMS)** that digitizes and automates core library operations, replacing the current paper-based workflow with a fast, accurate, and scalable digital platform.

2. Problem Statement

The existing paper-based library system suffers from the following challenges:

- Difficulty tracking book availability
- **Inefficient fine and overdue management**
- Difficulty tracking book availability
- Slow and error-prone record keeping

3. Motivation

The motivation for this project arises from the challenges associated with managing library operations using traditional manual methods. In the school library, activities such as student registration, borrowing and returning books, tracking overdue books, and maintaining user records are not done properly. These processes can be time-consuming and may increase the possibility of errors, misplaced records, and difficulties in determining the availability of books. With the library management system, all these activities will be easier to maintain and carry out.

4. Aims and Objective

The aim of this project is to **design and develop an efficient computerized Library Management System that automates and improves the management of library resources, users, book borrowing and returning, and library records, thereby reducing manual work, and improving the overall efficiency of library operations.**

Specific Objectives include:

- Design and develop a centralized system for managing library users, books, and library records.
- Implement a user registration and management module for maintaining accurate records of students and other authorized library users.
- Develop a book management module for recording and managing book details, including accession numbers, titles, authors, call numbers, categories, and availability status.

- Automate the book borrowing and returning process, including recording the borrower, date borrowed, and expected return date.
- Implement a book tracking mechanism that enables library staff to determine which books are borrowed, available, overdue, or returned.
- Implement an overdue management feature to identify books that have exceeded their expected return dates and assist library staff in monitoring outstanding books.

5. Methodology and Approach

Phase 1: Requirement Analysis

- Data collection: Gather information from staff, security personnel, and administrators.
- Needs Assessment: Identify essential features such as requests, approvals, logs, and reporting.

Phase 2: System Design

- Platform Architecture: Design the system components for user interface, logic, and database.
- UI/UX Design: Create simple pages for request submission, approval, and gate pass history.

Phase 3: Development

- Frontend Development: Build a responsive interface using React.js and Tailwind CSS.
- Backend & Data Management: Use SQL or Supabase to manage staff data, gate pass requests, approvals, and logs securely.

6. Expected Result

1. A functional computerized Library Management System will be developed to support and improve the management of library operations.
2. The system will provide an organized database of library users and books, making it easier for library staff to access and update records.
3. The system will enable efficient registration and management of students and library members, including the management of their library borrowing privileges.
4. The system will provide an effective book cataloguing and tracking mechanism using information such as accession numbers, call numbers, titles, authors, categories, and availability status.
5. The borrowing and returning of books will be properly recorded, allowing library staff to easily identify who borrowed a particular book and when it is due for return.